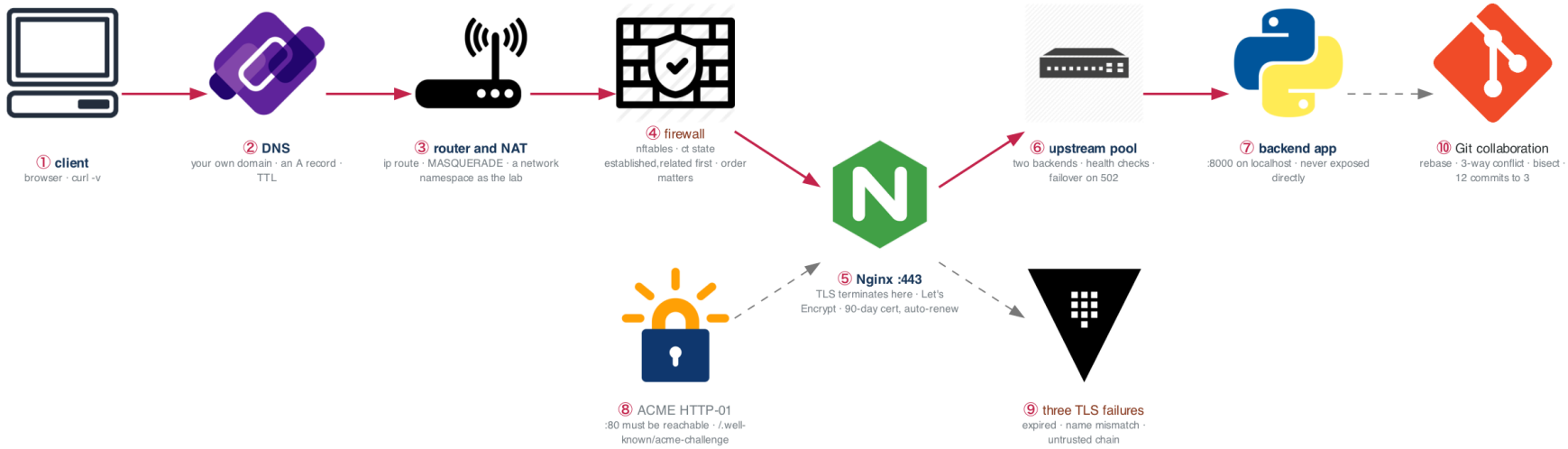
Browser to backend: routing, NAT, firewall, TLS, and the reverse proxy, numbered

Week 3 · Figure 3.1 · figure with badges 1–10 · 2026-10-17



What to say, badge by badge

About three and a half minutes at a normal pace (403 words). Point at each badge as you say its line.

Open
“This is the packet path you will be asked to explain at Gate 1: a browser on the internet to a Python process on your VM. Seven boxes on the blue line, each one built by hand this week.”

1 client
“Start with `curl -v`. It prints the DNS answer, the TCP connect, the TLS handshake, and the HTTP exchange, in order. When something breaks, the line where curl stops names the layer.”

2 DNS
“Your own domain, bought this week for about twelve dollars. One A record points at the VM. The TTL you choose is how long a wrong answer lives in every cache on the internet.”

3 router and NAT
“A packet that leaves a private address needs its source rewritten. MASQUERADE does that at the router, and the conntrack table remembers the mapping so the reply finds its way back. You build this with a network namespace, so you can see both sides.”

4 firewall
“nftables evaluates rules in order and stops at the first match. The stateful rule, `ct state established,related accept`, goes first; without it every reply is dropped. BF-13 is exactly this: a correct rule in the wrong position. Stateful versus stateless is the whole SG versus NACL conversation in week 6.”

5 Nginx on 443
“TLS terminates here. The certificate is from Let's Encrypt, valid 90 days, renewed by a timer. Nginx is the only process that listens on a public port.”

6 upstream pool
“Two backends behind one name. Health checks mark one down; traffic fails over on a 502. Kill a backend and watch the other take the load. That is a load balancer, and you now know what the ALB does.”

7 backend app
“The Python process listens on 8000, on localhost only. It never sees the internet. If it did, the firewall and the proxy would be decoration.”

8 ACME HTTP-01
“Let's Encrypt proves you own the name by fetching a token over port 80. If 80 is closed, issuance fails. That is the reason the firewall opens 80 even though users use 443.”

9 three TLS failures
“Expired, name mismatch, untrusted chain. They look identical in a browser and completely different in `openssl s_client`. Break each one on purpose so you recognize it on sight.”

10 Git collaboration
“Friday afternoon: resolve a real three-way conflict, squash twelve commits into three with an interactive rebase, and find a breaking commit with bisect in log-two-n steps.”

Close
“Name, route, rewrite, filter, terminate, balance, serve. Weeks 6 and 7 are this diagram with AWS names: Route 53, NAT gateway, security group, ALB, target group.”